Database Systems (CS-151)

WEEK-02

Course Instructor: Munazza

Course administration

Instructor: Munazza

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Consultation Hours 02:00PM - 03:00 PM

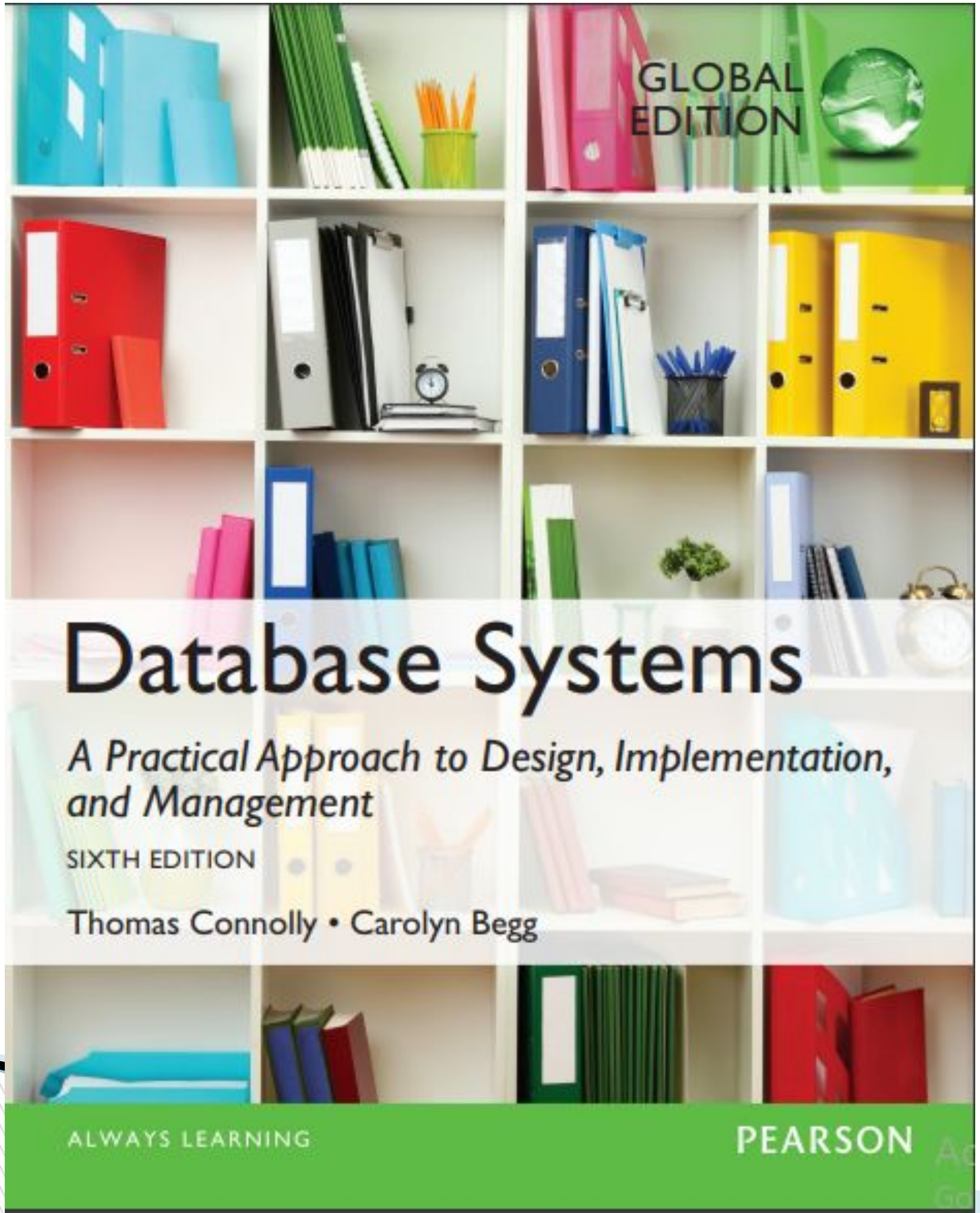
Course Learning Outcomes

Course Learning Outcomes		
After successful completion of this course, the students should be able to:	GA	BT Level
1. **Explain** fundamental database concepts	2	C2
2. **Identify** functional dependencies and **Produce** SQL queries for database definition and manipulation in any DBMS	3	C3
3. **Discover** conceptual, logical and physical database schemas using different data models	4	C4

Mapping of GA’s/PLO’s with CLO’s

Mapping of CLOs to GA’s

GA’s/CLOs	CLO1	CLO2	CLO3	CLO4
GA1 (Academic Education)				
GA2 (Knowledge for Solving Computing Problems)	✓			
GA3 (Problem Analysis)			✓	
GA4 (Design/Development of Solutions)		✓		
GA5 (Modern Tool Usage)				
GA6 (Individual and Teamwork)				
GA7 (Communication)				
GA8 (Computing Professionalism and Society)				
GA9 (Ethics)				
GA10 (Life-long Learning)				



Recommended Textbook and Reference Books

Text Book:

Database Systems: A Practical Approach to Design, Implementation, and Management by Thomas Connolly and Carolyn Begg 6th Edition, 2021

Text Book:

Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition 2002Database System Concepts by Avi Silberschatz, Henry F. Korth and S. Sudarshan, 7th Edition, 2019

Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Wido, 2nd Edition, 2008

Course Overview

- Introduction and significance of Databases
- Database Environment
- Database Languages
- Database Architectures and the web
- Relational model
- Database system development lifecycle
- Entity-relationship modelling
- Structural constraints
- Enhanced Entity-relationship modelling
- Normalization
- Functional dependencies
- SQL: Data Manipulation
- SQL Queries
- Ethical and security issues in database

Pre-requisites

- None

Grading criteria

Assessment	%Age
Assignments/ Quizzes	18.75%
Viva/Sessional	25%
Mid Term Exam	22.5%
Final Term Exam	33.75%
Total	100%

Goals for today

- To introduce:
  - Database
  - Advantages/Disadvantages of database

Database

- *“Database is an organized collection of related data stored in an efficient and compact manner”*.
  - **"organized"** means that data is stored in such a way that it can easily be accessed and updated
  - **"related data“** means data and information about a particular area is stored such as:
    - Database of employees contains data of employees of that particular organization or department
    - Database of students contains data of students of a college/university etc.
  - **"efficient“** means required data can be searched very easily and quickly
  - **"compact“** means that stored data takes up as little space as possible without any duplication of data

Examples of Databases

- NADRA
- Library
- College/University
- Bank Accounts
- E-mail Accounts
- Etc.

Types of databases

- Centralized databases

  Database whose complete data is located at a single computer such as Personal computer database, client-server database and central computer database.

- Distributed databases

  A single logical database which is spread physically across multiple computers over wide area network. Such databases are managed by the organizations having offices in different cities or countries.

Database Management System (DBMS)

- A collection of programs that are used to create, maintain, and extract data from databases
- It is general-purpose software
- This is often called database software
- Different types of DBMSs are available, ranging from small systems that run on personal computers to huge systems that run on mainframes

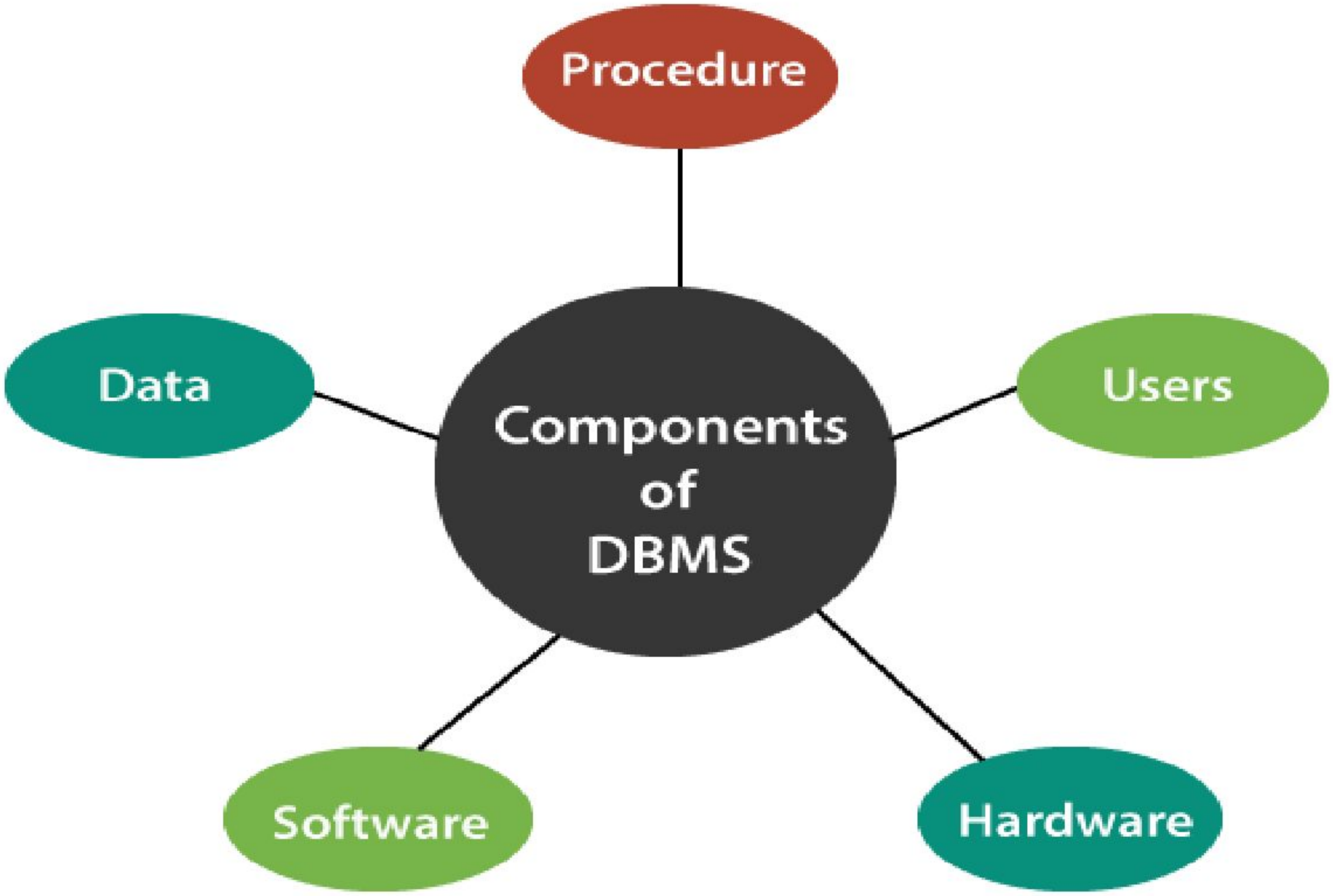
Functions of DBMS

- **Defining the Structure of Database**
  - It involves defining tables, fields and their data types, constraints for data to be stored in the database, and the relationships among tables.
- **Populating the Database**
  - Means storing data into database
- **Manipulating the Database**
  - In involves to retrieve specific data, update data, delete data, insert new data, and generate reports.

Examples of DBMS

- Microsoft Access
- Oracle
- Microsoft SQL Server
- MySQL
- FileMaker Pro

Components of a DBMS Environment



Components of a DBMS Environment

- **Hardware**
  - This component of DBMS consists of a set of physical electronic devices such as computers, I/O channels, storage devices, etc. that create an interface between computers and the users.
  - This DBMS component is used for keeping and storing the data in the database.

Components of a DBMS Environment

- **Software**
  - The main component of a DBMS is the software.
  - It is the set of programs which is used to manage the database and to control the overall computerized database.
  - The DBMS software provides an easy-to-use interface to store, retrieve, and update data in the database.
  - This software component is capable of understanding the Database Access Language and converts it into actual database commands to execute or run them on the database.

Components of a DBMS Environment

- **Data**
  - It is the most important component of the database management system.
  - In DBMS, databases are defined, constructed, and then data is stored, retrieved, and updated to and from the databases.
  - The database contains both the metadata (description about data or data about data) and the actual (or operational) data.

Components of a DBMS Environment

- **Procedures**
  - Procedures refer to general rules and instructions that help to design the database and to use the DBMS.
  - Procedures are used to;
    - Setup and install a new DBMS
      - to login and logout of DBMS software
      - to manage DBMS or application programs
      - to take backup of the database
      - to change the structure of the database, etc

Components of a DBMS Environment

- **Users**
  - The users are the people who control and manage the databases and perform different types of operations on the databases in the database management system.
  - There are three types of user who play different roles in DBMS:
    - Application Programmers
    - Database Administrators
    - End-Users

Different Users Role in DBMS

- **Application Programmers**
  - The users who write the application programs in programming languages (such as Java, C++, or Visual Basic) to interact with databases are called Application Programmer.

- **Database Administrators (DBA)**
  - A person who manages the overall DBMS is called a database administrator or simply DBA.

- **End-Users**
  - The end-users are those who interact with the database management system to perform different operations by using the different database commands such as insert, update, retrieve, and delete on the data, etc.

Advantages of Database

- **Controlling Data Redundancy**
  - In traditional file system, each application program has its own data files and this result in duplication of data in more than one places.
  - Database approach reduces data redundancy.
  - The data appears in a database appears only once and it is not duplicated.
  - Through controlling data redundancy storage space is saved.
    - Multiple files of same content with different name can be stored in file system
- **Data Consistency**
  - By controlling data redundancy, data consistency in obtained.
  - Data item is appeared only once
  - Any update in the value is performed only once and it is readily available to everyone.

Advantages of Database

- **Data Sharing**
  - Allows to share (access) data by any number of users simultaneously. (Rules: RR RW WW WR)
- **Data Integration**
  - In traditional file system, data is stored in separate files and it is difficult to extract the required information.
  - In database, data is stored in tables.
  - A single database may contain multiple tables.
  - The relationships can be created between tables .
  - This makes easier to retrieve and update data.

Advantages of Database

- **Data Integrity**
  - Data integrity refers to the correctness and consistency of data.
  - It is expressed in terms of certain constraints.
  - The consistency rules are applied on database to check that the correct data is entered in the database, before storing.
    - Ensures that each record in a table is unique and identifiable. This is typically achieved by using a **Primary Key**, which is a unique identifier for a record.

Advantages of Database

- **Data Atomicity**
  - Atomicity means that either one transaction should take place as a whole or it should not take place at all.
  - If the any process is not completed successfully then the system fails and this is known as data atomicity problem.
  - In database partially completed tasks are rolled back.
  - Only consistent data exists within the database

Data Atomicity

- Banking system with two accounts: Account A and Account B. If a user transfers money from Account A to Account B, the system must ensure that both operations (debit from Account A and credit to Account B) are performed atomically.
- **Transaction Example**: Transfer $100 from Account A to Account B.
  - Debit $100 from Account A.
  - Credit $100 to Account B.
- If the transaction is atomic:
  - If both operations are successful, the transaction is committed, and the balances are updated.
- If an error occurs after the debit from Account A but before the credit to Account B, atomicity ensures the transaction is rolled back. In this case, the $100 is not deducted from Account A, and no money is added to Account B.

Advantages of Database

- **Data Security**
  - It is the protection of the database from unauthorized users.
  - Only authorized persons are allowed to access the database.
    - (Role based access control/ Hierarchal Security)

- **Control Over Concurrency**
  - In some situations two or more users may access the same file simultaneously.
  - It is possible that they will interfere with each other.
    - For example, two or more users are trying to update the same record, then one may overwrite the values recorded by other user. This may result in loss of information.
  - Most databases control the concurrency so that transactions are recorded with accuracy.

Advantages of Database

9. **Better Backup and Recovery Facility:**

    - Data is a very important resource and is very much valuable for any organization, loss of such a valuable resource can result in a huge strategic disasters.
    - As Data is stored on today's' storage devices like hard disks etc., It is necessary to take periodic backups of data so that in case a storage device looses the data due to any damage we should be able to restore the data at nearest point.
    - Database systems offer excellent facilities for taking backup of data and good mechanism of restoring those backups to get back the backed-up data.

Advantages of Database

- **Data Independence**
  - The data stored in a database is independent of the application programs that accesses the data from database.
  - The data structure of database and application program that uses data are separate from each other.
  - The user can easily change the structure of database without modifying the application program.
  - Similarly, user can modify application programs without changing structure of database.

Disadvantages of Database

1. **High Cost:**

Database includes the need for specialized **software** which is used to run database systems, Additional and specialized **hardware** and technically **qualified staff** are the requirements for adopting to the database system, all these requirements need an organization to invest handsome amount of money to have all the requirements of the database systems.

Disadvantages of Database

2. **Conversion Cost:**

Once an organization has decided to adopt database system for its operations, it is not only the finance and technical man-power which is required for switching on to database system, it further has some conversion charges needed for adopting the database system, this is also a very important stage for making decision about the way the system will be converted to database system.

Disadvantages of Database

3. **Difficult Recovery Procedures:**

Although the database systems and database management systems provide very efficient ways of data recovery in case of any disaster, still the process of recovering a crashed database is very much technical and needs good professional skills to perform a perfect recovery of the database.